

LECTURE 5

4.3. Linear non-homogeneous recurrence relations with constant coefficients. Using characteristic equations we have solved

$$(4.3) \quad a_{n+k} + c_1 a_{n+k-1} + c_2 a_{n+k-2} + \cdots + c_k a_n = g(n)$$

in the case $g(n) \equiv 0$ (the homogeneous case) for $k = 1, 2$. (The same method works for larger k as well.)

What about $g(n) \not\equiv 0$? The equation (4.3) is linear so if $a_n^{(h)}$ are all solutions to the homogeneous equation it is easy to show that all solutions to (4.3) are given by $a_n = a_n^{(h)} + a_n^{(p)}$ where $a_n^{(p)}$ is any solution to (4.3). Indeed, since $a_n^{(p)}$ is a solution to (4.3), we have

$$a_{n+k}^{(p)} + c_1 a_{n+k-1}^{(p)} + c_2 a_{n+k-2}^{(p)} + \cdots + c_k a_n^{(p)} = g(n).$$

Then substituting $a_n = a_n^{(h)} + a_n^{(p)}$ into the left-hand side of (4.3) yields

$$\begin{aligned} & (a_{n+k}^{(h)} + a_{n+k}^{(p)}) + c_1(a_{n+k-1}^{(h)} + a_{n+k-1}^{(p)}) + c_2(a_{n+k-2}^{(h)} + a_{n+k-2}^{(p)}) + \cdots + c_k(a_n^{(h)} + a_n^{(p)}) \\ &= (a_{n+k}^{(h)} + c_1 a_{n+k-1}^{(h)} + \cdots + c_k a_n^{(h)}) + (a_{n+k}^{(p)} + c_1 a_{n+k-1}^{(p)} + \cdots + c_k a_n^{(p)}) \\ &= 0 + g(n) \\ &= g(n). \end{aligned}$$

How can we find $a_n^{(p)}$, a particular solution? In most cases $a_n^{(p)}$ are of the same form of $g(n)$.

$g(n)$	$a_n^{(p)}$
polynomial of degree k	polynomial of degree k
$C r^n$	$A r^n$
$C \sin n\theta + D \cos n\theta$	$A \sin n\theta + B \cos n\theta$
r^n (poly. of deg. k)	r^n (poly. of deg. k)
$r^n(C \sin n\theta + D \cos n\theta)$	$r^n(A \sin n\theta + B \cos n\theta)$

Example 4.11. Find all solutions to

$$(4.4) \quad a_{n+2} - a_{n+1} - 2a_n = n^2 - 3n + 1.$$

Solution: We first solve the homogenous equation $a_{n+2} - a_{n+1} - 2a_n = 0$. The characteristic equation is $r^2 - r - 2 = 0$, which has the solutions $r = \frac{1}{2} \pm \sqrt{\frac{1}{4} + 2} = \frac{1}{2} \pm \frac{3}{2} = 2, -1$. Hence $a_n^{(h)} = A \cdot 2^n + B(-1)^n$.

To find $a_n^{(p)}$: set $a_n^{(p)} = Cn^2 + Dn + E$ (same form as RHS) and plug it into (4.4):

$$C(n+1)^2 + D(n+2) + E - C(n+1)^2 - D(n+1) - E - 2Cn^2 - 2Dn - 2E = n^2 - 3n + 1$$

$\iff$

$$\begin{aligned} n^2(C - C - 2C) + n(4C + D - 2C - D - 2D) + (4C + 2D + E - C - D - E - 2E) \\ = n^2 - 3n + 1 \end{aligned}$$

$$\iff \begin{cases} -2C = 1 & (n^2) \\ 2C - 2D = -3 & (n) \\ 3C + D - 2E = 1 & (\text{constant}) \end{cases}$$

$$\iff \begin{cases} C = -\frac{1}{2} \\ D = \frac{-3-2C}{-2} = 1 \\ E = \frac{1-3C-D}{-2} = \frac{1+\frac{3}{2}-1}{-2} = -\frac{3}{4} \end{cases}$$

Therefore $a_n^{(p)} = -\frac{n^2}{2} + n - \frac{3}{4}$ and hence (4.4) has the solutions

$$\begin{aligned} a_n &= a_n^{(h)} + a_n^{(p)} \\ &= A \cdot 2^n + B(-1)^n - \frac{n^2}{2} + n - \frac{3}{4}. \end{aligned}$$

Example 4.12. Find all solutions for $a_{n+1} - a_{n+1} - 2a_n = 5 \cdot 3^n$.

Solution: As in the previous example $a_n^{(h)} = A \cdot 2^n + B(-1)^n$. Here we try with $a_n^{(p)} = C \cdot 3^n$. Plugging in gives:

$$\begin{aligned} C \cdot 3^{n+2} - C \cdot 3^{n+1} - 2C \cdot 3^n &= 5 \cdot 3^n & \iff & 9C - 3C - 2C = 5 \\ & & \iff & 4C = 5 \\ & & \iff & C = \frac{5}{4} \end{aligned}$$

Hence $a_n^{(p)} = \frac{5}{4} \cdot 3^n$ is a solution and all solutions are given by

$$a_n = a_n^{(h)} + a_n^{(p)} = A \cdot 2^n + B(-1)^n + \frac{5}{4} \cdot 3^n.$$

Sometimes the above method fails to find a particular solution.

Example 4.13. Find a solution to $a_{n+2} - a_{n+1} - 2a_n = 5 \cdot 2^n$.

Solution: Trying with $a_n^{(p)} = C \cdot 2^n$ gives $0 = 5 \cdot 2^n$ because $a_n^{(p)} = C \cdot 2^n$ solves the homogenous equation.

In such a case we use n times the solution we would normal try. Here $a_n^{(p)} = C \cdot n \cdot 2^n$ which results in

$$\begin{aligned} C(n+2)2^{n+2} - C(n+1)2^{n+1} - 2C \cdot n \cdot 2^n &= 5 \cdot 2^n & \iff & 4(C(n+2) - 2C(n+1) - 2Cn) = 5 \\ & & \iff & 8C - 2C = 5 \\ & & \iff & C = \frac{5}{6}. \end{aligned}$$

Thus we know that $a_n^{(p)} = \frac{5}{6}n \cdot 2^n$ would be a solution.

If some term of the $a_n^{(p)}$ that we would normally try solves the homogenous equation then multiply $a_n^{(p)}$ by n . (One may need to repeat, for example the homogenous equation has the solution $A \cdot 4^n + B \cdot n \cdot 4^n$ (double root case) and RHS equals $18 \cdot 4^n$. This implies that $a_n^{(p)} = C \cdot 4^n$ will not work, hence try $a_n^{(p)} = C \cdot n \cdot 4^n$. This will not work either because it still solves the homogenous equation. Therefore one should try $a_n^{(p)} = C \cdot n^2 \cdot 4^n$. This will work.)

4.4. Generating functions to solve recurrence relations.

Example 4.14. Solve

$$(4.5) \quad a_{n+2} - 3a_{n+1} + 2a_n = 0$$

for all $n \geq 0$ with $a_0 = 1$, $a_1 = 6$, using generating functions.

Solution: Let $F(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots$. If we can find $F(x)$ and extract its coefficient of x^n , then we have found a_n .

Multiply (4.5) by x^{n+2} :

$$a_{n+2}x^{n+2} - 3a_{n+1}x^{n+2} + 2a_nx^{n+2} = 0.$$

Now summing over n from 0 to ∞ gives

$$\sum_{n=0}^{\infty} a_{n+2}x^{n+2} - 3x \sum_{n=0}^{\infty} a_{n+1}x^{n+1} + 2x^2 \sum_{n=0}^{\infty} a_nx^n = 0.$$

We observe that

$$\begin{aligned} \sum_{n=0}^{\infty} a_{n+2}x^{n+2} &= F(x) - a_0 - a_1x = F(x) - 1 - 6x, \\ \sum_{n=0}^{\infty} a_{n+1}x^{n+1} &= F(x) - a_0 = F(x) - 1, \\ \sum_{n=0}^{\infty} a_nx^n &= F(x). \end{aligned}$$

So we have found that

$$F(x) - 1 - 6x - 3x(F(x) - 1) + 2x^2F(x) = 0$$

$$\iff (1 - 3x + 2x^2)F(x) = 1 + 3x$$

$$\begin{aligned} \text{Partial fractions decomposition} \iff F(x) &= \frac{1 + 3x}{1 - 3x + 2x^2} = \frac{1 + 3x}{2(1 - x)(\frac{1}{2} - x)} = \frac{A}{1 - x} + \frac{B}{\frac{1}{2} - x} \\ &= \frac{[A(\frac{1}{2} - x) + B(1 - x)]2}{2(1 - x)(\frac{1}{2} - x)} \end{aligned}$$

$$\iff 1 + 3x = A + 2B + x(-2A - 2B)$$

$$\iff \begin{cases} A + 2B = 1 \\ -2A - 2B = 3 \end{cases}$$

$$\iff \begin{cases} A = -4, \\ B = \frac{1-A}{2} = \frac{5}{2}. \end{cases}$$

This shows that

$$F(x) = \frac{-4}{1-x} + \frac{5}{2} \frac{1}{(\frac{1}{2}-x)} = \frac{-4}{1-x} + \frac{5}{1-2x} = -4 \sum_{j=0}^{\infty} x^j + 5 \sum_{j=0}^{\infty} (2x)^j.$$

The coefficient of x^n is $-4 + 5 \cdot 2^n$, so the answer is $a_n = -4 + 5 \cdot 2^n$.

Example 4.15. Solve

$$(4.6) \quad a_n - 3a_{n-1} = n^2$$

for all $n \geq 1$ with $a_0 = 1$, using generating functions.

Solution: Let $F(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots$. As before, the coefficient of x^n gives us a_n .

Multiply (4.6) by x^n and summing over n from 1 to ∞ gives:

$$\sum_{n=1}^{\infty} a_n x^n - 3x \sum_{n=1}^{\infty} a_{n-1} x^{n-1} = \sum_{n=1}^{\infty} n^2 x^n$$

Note that

$$\sum_{n=1}^{\infty} a_n x^n = F(x) - 1 \quad \text{and} \quad \sum_{n=1}^{\infty} a_{n-1} x^{n-1} = F(x),$$

so, recalling from Example 3.14 that

$$\sum_{n=1}^{\infty} n^2 x^n = \frac{x(1+x)}{(1-x)^3},$$

we obtain

$$F(x) - 1 - 3xF(x) = \frac{x(1+x)}{(1-x)^3}.$$

Rearranging yields

$$\begin{aligned} F(x) &= \frac{x^2 + x}{(1-x)^3(1-3x)} + \frac{1}{(1-3x)} \\ &= \left[\frac{A}{1-x} + \frac{B}{(1-x)^2} + \frac{C}{(1-x)^3} + \frac{D}{1-3x} \right] + \frac{1}{1-3x} \end{aligned} \quad (\text{Partial fractions})$$

and therefore,

$$\begin{aligned} x^2 + x &= A(1-x)^2(1-3x) + B(1-x)(1-3x) + C(1-3x) + D(1-x)^3 \\ &= A(1-5x+7x^2-3x^3) + B(1-4x+3x^2) + C(1-3x) + D(1-3x+3x^2-x^3). \end{aligned}$$

From comparing coefficients of x^3 , x^2 , x^1 and x^0 , we get

$$\begin{aligned}
 \begin{cases} -3A - D = 0 \\ 7A + 3B + 3D = 1 \\ -5A - 4B - 3C - 3D = 1 \\ A + B + C + D = 0 \end{cases} &\iff \begin{cases} D = -3A \\ 7A + 3B + 3(-3A) = 1 \\ -5A - 4B - 3C - 3(-3A) = 1 \\ A + B + C - 3A = 0 \end{cases} \\
 &\iff \begin{cases} D = -3A \\ -2A + 3B = 1 \\ 4A - 4B - 3C = 1 \\ -2A + B + C = 0 \end{cases} \\
 &\iff \begin{cases} D = -3A \\ B = \frac{1+2A}{3} \\ 12A - 4(1+2A) - 9C = 3 \\ -6A + (1+2A) + 3C = 0 \end{cases} \\
 &\iff \begin{cases} D = -3A \\ B = \frac{1+2A}{3} \\ 4A - 9C = 7 \\ -4A + 3C = -1 \end{cases} \\
 &\iff \begin{cases} A = -\frac{1}{2}, \\ B = 0, \\ C = -1, \\ D = \frac{3}{2}. \end{cases}
 \end{aligned}$$

This shows that

$$F(x) = -\frac{1}{2(1-x)} - \frac{1}{(1-x)^3} + \frac{5}{2(1-3x)} = -\frac{1}{2} \sum_{j=0}^{\infty} x^j - \sum_{j=0}^{\infty} \binom{-3}{j} x^j + \frac{5}{2} \sum_{j=0}^{\infty} (3x)^j.$$

The coefficient of x^n is $-\frac{1}{2} - \binom{-3}{n} + \frac{5}{2} \cdot 3^n$, so the answer is $a_n = -\frac{1}{2} - \binom{-3}{n} + \frac{5}{2} \cdot 3^n$.